

Non deterministic Finite Automata (NFA)

$$N = (Q, \Sigma, \Delta, S, F)$$

Q - Finite set of States.

$S \subseteq Q$ - set of start states $F \subseteq Q$ set of final states.

$$\Delta: Q \times \Sigma \rightarrow 2^Q \quad \text{where } 2^Q = \{A \mid A \subseteq Q\}.$$

$\Delta(q, a)$ - set of all states that N is allowed to move to from q in one step under the symbol a .

$$q \xrightarrow{a} p \quad \text{if } p \in \underbrace{\Delta(q, a)}_{\text{can be empty.}}$$

Extending Δ over strings.

$$\hat{\Delta}: 2^Q \times \Sigma^* \rightarrow 2^Q$$

$$\hat{\Delta}(A, \epsilon) = A.$$

$$\hat{\Delta}(A, xa) = \bigcup_{q \in \hat{\Delta}(A, x)} \Delta(q, a)$$

For $A \subseteq Q$ and $x \in \Sigma^*$, $\hat{\Delta}(A, x)$ is the set of all states reachable under string x from some state in A .

$$r \in \hat{\Delta}(A, xa) \text{ if } \exists q \in \hat{\Delta}(A, x) \text{ and } r \in \Delta(q, a)$$

$$p \xrightarrow{x} q \xrightarrow{a} r$$

$$\hat{\Delta}(A, a) = \bigcup_{p \in \hat{\Delta}(A, \epsilon)} \Delta(p, a) = \bigcup_{p \in A} \Delta(p, a)$$

N accepts $x \in \Sigma^*$ if $\hat{\Delta}(S, x) \cap F \neq \emptyset$.

$$L(N) = \{x \in \Sigma^* \mid N \text{ accepts } x\}.$$

DFA $M = (Q, \Sigma, S, \delta, F)$ and NFA $N = (Q, \Sigma, \Delta, S, F)$

Statement. NFAs and DFAs have the same expressive power

Note. A DFA M can be easily "converted" to an equivalent NFA.

Take $S = \{\delta\}$ and $\Delta(q, a) = \{\delta(q, a)\}$

Subset Construction - Converts an NFA to an equivalent DFA.

Let $N = (Q_N, \Sigma, \Delta_N, S_N, F_N)$ to construct

$M = (Q_M, \Sigma, S_M, \delta_M, F_M)$ s.t. $L(M) = L(N)$.

$Q_M = 2^{Q_N}$ - the power set of Q_N .

$S_M(A, a) = \hat{\Delta}_N(A, a)$

$\delta_M = S_N$

$F_M = \{A \subseteq Q_N \mid A \cap F_N \neq \emptyset\}$.

Lemma 1. For any $x, y \in \Sigma^*$, and $A \in Q$.

$$\hat{\Delta}(A, xy) = \hat{\Delta}(\hat{\Delta}(A, x), y).$$

[Proof by induction on $|y|$]

Lemma 2. For any $A \in Q_N$, and $x \in \Sigma^*$

$$\hat{S}_M(A, x) = \hat{\Delta}_N(A, x)$$

Proof. By induction on $|x|$

Base Case: $x = \epsilon$ $\hat{S}_M(A, \epsilon) = A$ $\hat{\Delta}_N(A, \epsilon) = A$.



Induction Step.

$$\hat{S}_M(A, xa) = S_M(\hat{S}_M(A, x), a) \text{ [defn. of } \hat{S}_M]$$

$$= S_M(\hat{\Delta}_N(A, x), a) \text{ [Induction Hypothesis]}$$

$$= \hat{\Delta}_N(\hat{\Delta}_N(A, x), a) \text{ [Defn of } S_M]$$

$$= \hat{\Delta}_N(A, xa) \text{ [Lemma 1.]}$$

Lemma 1. For any $x, y \in \Sigma^*$, and $A \in Q$.

$$\hat{\Delta}(A, xy) = \hat{\Delta}(\hat{\Delta}(A, x), y).$$

[Proof by induction on $|y|$]

Lemma 2. For any $A \in Q_N$, and $x \in \Sigma^*$

$$\hat{S}_M(A, x) = \hat{\Delta}_N(A, x)$$

Proof. By induction on $|x|$

Base Case: $x = \epsilon$ $\hat{S}_M(A, \epsilon) = A$ $\hat{\Delta}_N(A, \epsilon) = A$.



Induction Step.

$$\begin{aligned} \hat{S}_M(A, xa) &= S_M(\hat{S}_M(A, x), a) \quad [\text{defn. of } \hat{S}_M] \\ &= S_M(\hat{\Delta}_N(A, x), a) \quad [\text{Induction Hypothesis}] \\ &= \hat{\Delta}_N(\hat{\Delta}_N(A, x), a) \quad [\text{Defn of } S_M] \\ &= \hat{\Delta}_N(A, xa) \quad [\text{Lemma 1.}] \end{aligned}$$

Theorem. $L(M) = L(N)$

For $x \in \Sigma^*$ $x \in L(M) \Leftrightarrow \hat{S}_M(S_M, x) \in F_M$ [acceptance]

$$\Leftrightarrow \hat{\Delta}_N(S_N, x) \cap F_N \neq \emptyset \quad [\text{Lemma 2, } S_M, F_M]$$

$$\Leftrightarrow x \in L(N) \quad [\text{Defn of acceptance of } N]$$

DFA $M = (Q, \Sigma, S, \delta, F)$
 $S: Q \times \Sigma \rightarrow Q \quad \delta \in Q.$

$$L(M) = \{x \in \Sigma^* \mid M \text{ accepts } x\}$$

NFA $N = (Q, \Sigma, \Delta, S, F)$

$$\Delta: Q \times \Sigma \rightarrow 2^Q \quad ; \quad S \subseteq Q.$$

$$L(N) = \{x \in \Sigma^* \mid N \text{ accepts } x\}.$$

Regular sets.

$A \subseteq \Sigma^*$ is regular if $\exists$ a DFA M s.t. $L(M) = A$.

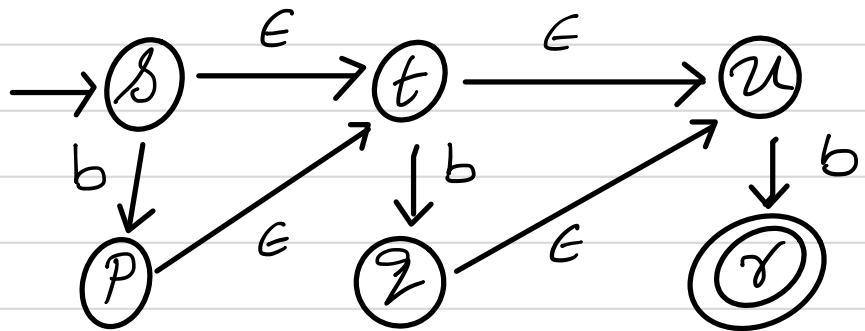
For every NFA N $\exists$ a DFA M s.t. $L(M) = L(N)$

$A \subseteq \Sigma^*$ is regular if $\exists$ a NFA s.t. $L(N) = A$.

NFA with ϵ -transitions

$p \xrightarrow{\epsilon} q$: The automaton can take an ϵ -transition any time without reading an input symbol

Example - N:



If N is in state s and the next symbol is b then M can do the following:

- Read b and move to state p .
- Move to t without reading any symbol and then read b and move to q .
- Move to t (on ϵ), move to u (on ϵ), read b and move to r .

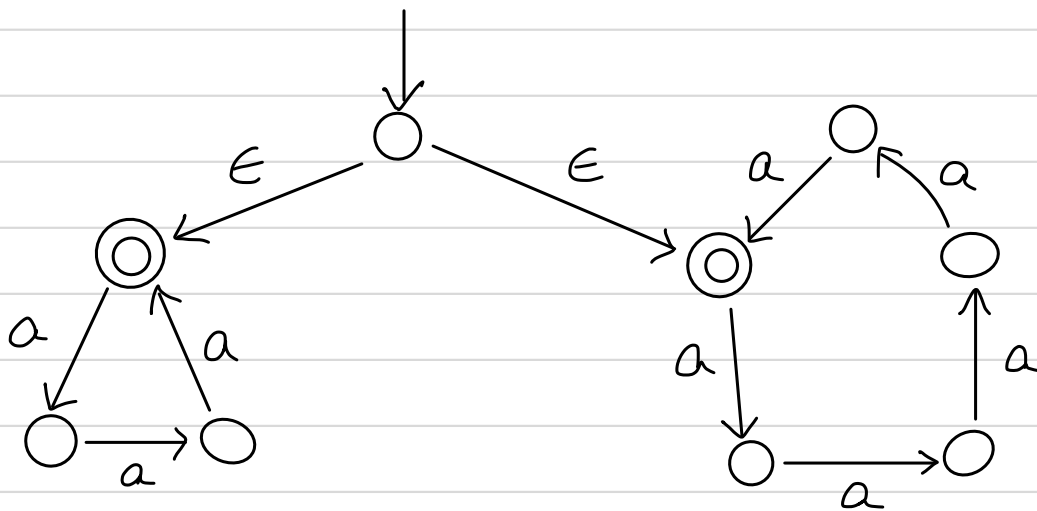
$$L(N) = \{b, bb, bbb\}$$

Example $\Sigma = \{a\}$

$$A = \{w \in \Sigma^* \mid |w| \text{ is divisible by } 3\}$$

$$B = \{w \in \Sigma^* \mid |w| \text{ is divisible by } 5\}$$

$$C = \{w \in \Sigma^* \mid |w| \text{ is divisible by } 3 \text{ or } 5\}$$



Theorem. For every ϵ -NFA N , there exists a DFA M s.t. $L(M) = L(N)$.

Conclusion: ϵ -transitions do not add expressive power, it adds convenience.

Closure properties (Revisited)

- Regular sets are closed under intersection & union (the product construction).

Union

- De Morgan's Law
- Product construction
- A simpler construction using an NFA?

